Article

AERIS: Environment-Aware Hierarchical Routing for Reliable Wireless Sensor Networks under Realistic Channel Conditions

Kangrui Li [1,*], Xiaobo Zhang [1] and Junyi Lin [1]

[1] Faculty of Automation, Guangdong University of Technology, Guangzhou 510006, China
* Correspondence: 1403073295@mails.gdut.edu.cn

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Abstract: This study evaluates AERIS, a lightweight hierarchical routing protocol for wireless sensor networks under heterogeneous channel conditions, using `pdr_expected` as the primary metric. In the publication-tier 100-node regime (four environments, 30 seeds per environment), AERIS yields the highest mean PDR within the tested baseline set (LEACH, PEGASIS, HEED, and TEEN). In the frozen S8 scalability matrix (100–1000 nodes, four environments, 1000 replicates per environment-node-protocol cell), AERIS remains the highest-mean protocol in all 24 environment-scale cells, but this block is treated as bounded baseline evidence because MAC-collision modeling is absent and non-physical upward trends appear in three environments. We therefore use S9/S10/S11 as calibration blocks under stricter assumptions. S10 (tx-power 5 vs 15 dBm, n=600 per cell) shows significant differences in 59/60 cells after Holm correction (single non-significant case: LEACH at indoor_office, 1000 nodes). In the fully matched S11 patch-versus-control matrix (n=1000 in both arms), AERIS patch-control deltas are negative and significant in all 24 environment-scale cells, indicating reliability degradation under the current collision/relay patch configuration. NS-3 is used as trend-level cross-platform validation (AERIS vs LEACH), not as a numerical-equivalence proof.

Keywords: wireless sensor networks; reliable routing; hierarchical protocol; packet delivery ratio; scalability; reproducible evaluation

1. Introduction

Wireless sensor network (WSN) routing is classically optimized for energy reduction, but practical deployments also require stable delivery under heterogeneous channels [1–3]. This creates a multi-objective tension among reliability, hop-based latency, and energy use. Existing protocols represent distinct operating points: LEACH emphasizes clustered forwarding [4], PEGASIS emphasizes chain-based aggregation [5], HEED emphasizes residual-energy-aware head selection [6], and TEEN emphasizes event-triggered reporting [7]. Prior channel studies also show that low-power links can be asymmetric and environment-sensitive under realistic propagation conditions [8–10].

AERIS is designed as a lightweight hierarchical protocol that prioritizes reliability under realistic channel variability. The design objective is not universal dominance; rather, it is robust performance across multiple channel environments while preserving deployability on resource-constrained nodes.

Contributions and Scope Boundaries

This manuscript makes three scoped contributions:

- publication-tier evidence for multi-environment reliability at 100 nodes, plus balanced scalability analysis from 100 to 1000 nodes;

- module-level ablation evidence under the same channel taxonomy;
- cross-platform NS-3 trend validation for AERIS versus LEACH.

The manuscript focuses on reproducible conclusions under explicit settings (metric, environment taxonomy, baseline set, and sample-size design) and does not claim cross-platform numerical equivalence.

2. Related Work

Classical WSN protocols (LEACH, PEGASIS, HEED, TEEN) provide foundational routing mechanisms but show different failure modes under harsh channels and larger scales [3,5,6,11]. Chain-based aggregation can reduce per-hop transmission effort but increases end-to-end hop count. Cluster-based approaches can reduce hop count but may become sensitive to uplink quality and head placement.

Recent adaptive or learning-based methods improve specific settings but often require heavier runtime or training overhead [12–14]. Recent Sensors/IoT studies also report robust gains from context-aware or hybrid routing under constrained networks [15–17]. These results motivate explicit environmental adaptation in protocol design while keeping inference and control logic lightweight. In this manuscript, the focus is a rule-based protocol with explicit reproducibility constraints and auditable experiment metadata.

3. System Model and Protocol

3.1. Metric Definition

The primary metric is

$$\mathrm{PDR}_{\mathrm{expected}} = \frac{\text{bs\_delivered}}{\text{source\_packets\_expected}}. \tag{1}$$

All publication conclusions in this draft are based on this metric.

3.2. AERIS Architecture

AERIS combines three components:

1. Context-Adaptive Switching (CAS) for mode control,
2. Skeleton backbone for hierarchical forwarding,
3. Gateway coordination for CH-to-BS reinforcement.

Gateway candidate scoring follows a weighted rule

$$\mathrm{Score}_i = \alpha E_i + \beta C_i + \gamma L_i \tag{2}$$

where E_i is residual energy, C_i is centrality, and L_i is link quality.

3.3. Round-Level Operation

Each simulation round follows a fixed sequence: (i) cluster-head election and clustering, (ii) intra-cluster delivery with CAS mode selection, (iii) CH-to-BS forwarding with optional Gateway/Skeleton support, and (iv) metric logging. This ordering is held constant across protocols to preserve comparability in the reported matrices.

4. Experimental Setup

4.1. Publication-Tier Settings

- Simulator: Python implementation with logged provenance.

- Channel environments: indoor_office, indoor_factory, outdoor_urban, outdoor_suburban.
- Default node count: 100.
- Seeds for multi-environment and ablation: 30 independent seeds.
- Seeds for scalability: balanced frozen regime with n=1000 per environment-node-protocol cell (100–1000 nodes, 4 environments).
- Baselines: LEACH, PEGASIS, HEED, TEEN.

4.2. Evidence Scope

The analysis is grounded in publication-tier multi-environment comparison, ablation, scalability, hop-based latency, energy-lifetime statistics, and NS-3 trend validation. Cross-regime numerical pooling is avoided.

4.3. Simulator Regimes and Physics Assumptions

This study reports two simulation regimes for scalability interpretation. The frozen S8 regime is the original simulator path used for the balanced 100–1000 node matrix. The S9/S10/S11 regime introduces stricter physics controls (collision/relay stress, matched patch-control checks, and tx-power sensitivity) and is used for calibration and boundary checks. We therefore do not pool S8 and S9/S10/S11 values into a single headline estimate. Instead, S8 is treated as the baseline reporting matrix, while S9/S10/S11 are treated as rigor-calibration blocks that quantify how conclusions shift under stricter assumptions.

Table 1. Evidence-regime map used in this manuscript.

Block	Scope	Typical sample size	Role in claims
100-node matrix	4 env × 5 protocols	n=30 per env-protocol cell	Core reliability comparison
S8 scalability	4 env × 6 node counts × 5 protocols	n=1000 per env-node-protocol cell	Baseline large-scale matrix
S9 patch-control	4 env × 6 node counts × 5 protocols	patch n=1000, control n=600	Physics-stress calibration
S10 power sensitivity	4 env × 3 node counts × 5 protocols	n=600 per env-node-protocol cell	Tx-power boundary analysis
S11 matched patch-control	4 env × 6 node counts × 5 protocols	patch n=1000, control n=1000	Matched strict-physics confirmation
NS-3 alignment	4 env × 7 node counts, AERIS vs LEACH	n=30 per env-node-protocol cell	Trend-level cross-platform check

4.4. Statistical Methods

For pairwise protocol comparisons, we use Welch's two-sample t-test because variance equality is not assumed across protocols and environments. Family-wise error control is handled by Holm correction within each reported comparison family. Effect size is reported with Hedges' g and interpreted as small (0.2), medium (0.5), and large (0.8) in absolute value. Confidence intervals are reported as two-sided 95% intervals. All significance statements in this manuscript are based on Holm-corrected p values. For large-n cells, interpretation is based jointly on corrected p values, effect size, and absolute PDR delta. Reported table-wise standard deviations are treated as descriptive values from the corresponding result files; inferential conclusions are computed from raw replicate-level records.

4.5. Reproducibility Controls

All publication-tier scripts in this study use deterministic seed lists. They also enforce `force_ctp_reliable=False`. This guarantees that PDR reflects simulated delivery behavior rather than any reliability override. We report `pdr_expected` as the primary metric and treat hop count as a latency proxy, not a wall-clock latency measurement. Significance statements are accepted only when consistent with the declared regime-specific sample size.

5. Results

5.1. Multi-Environment Comparison at 100 Nodes (n=30)

Table 2. PDR by environment at 100 nodes (mean ± std; standard deviations are reported as stored in the source result file).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN
indoor_office	0.9739±0.0047	0.5543±0.0401	0.9078±0.0166	0.9371±0.0076	0.8222±0.0044
indoor_factory	0.6031±0.0258	0.1614±0.0209	0.1928±0.0255	0.2326±0.0263	0.3113±0.0245
outdoor_urban	0.3745±0.0354	0.0552±0.0127	0.0542±0.0117	0.0635±0.0121	0.1201±0.0183
outdoor_suburban	0.7451±0.0193	0.2703±0.0272	0.3382±0.0329	0.4221±0.0313	0.4752±0.0236

At 100 nodes, AERIS ranks first in all four tested environments within the evaluated baseline set (LEACH, PEGASIS, HEED, and TEEN). This claim is restricted to the 100-node matrix and should not be generalized to the 1000-node regime.

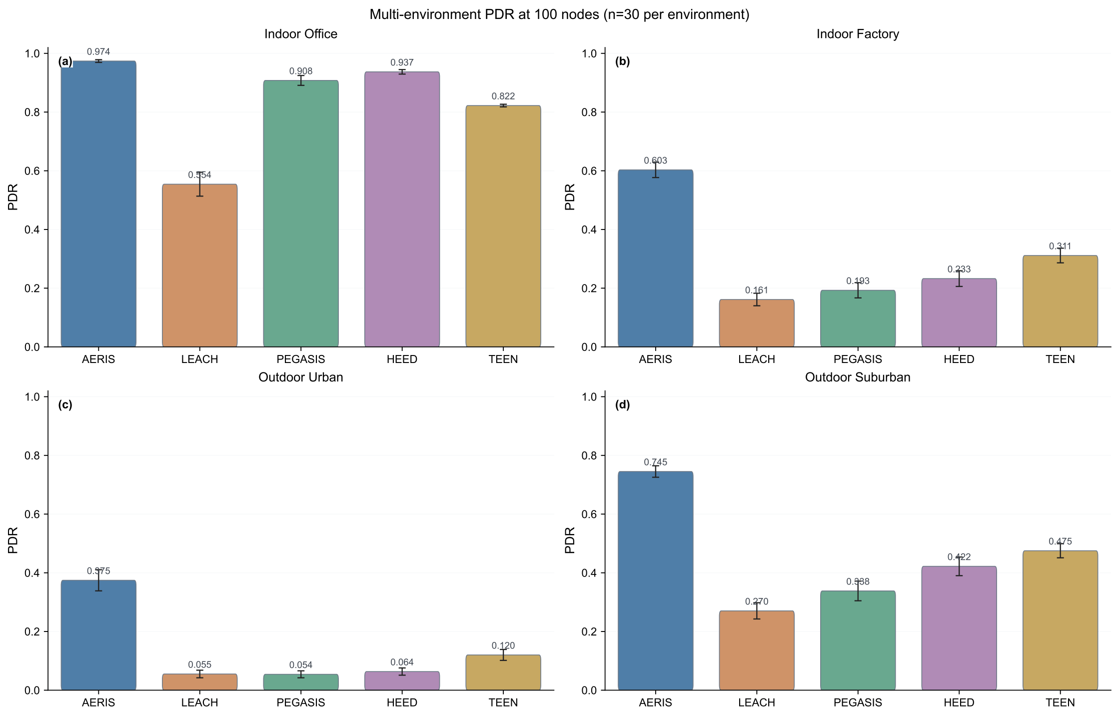

Figure 1. PDR comparison panel at 100 nodes (4 environments, n=30).

5.2. Ablation: Environment-Dependent Effects (n=30)

Table 3. Full vs no_gateway PDR (mean ± std; standard deviations are reported as stored in the source result file).

Environment	Full	no_gateway	Delta (no_gateway - full)
indoor_office	0.9739±0.0047	0.9741±0.0036	+0.0002
indoor_factory	0.6031±0.0258	0.5806±0.0215	-0.0225
outdoor_urban	0.3745±0.0354	0.3534±0.0301	-0.0212
outdoor_suburban	0.7451±0.0193	0.7306±0.0264	-0.0146

Gateway is positive in three environments and near-neutral in indoor_office. CAS is not consistently positive across environments. In indoor_office, the Full-vs-no_gateway gap is +0.0002 (absolute PDR), so this cell is treated as practically neutral.

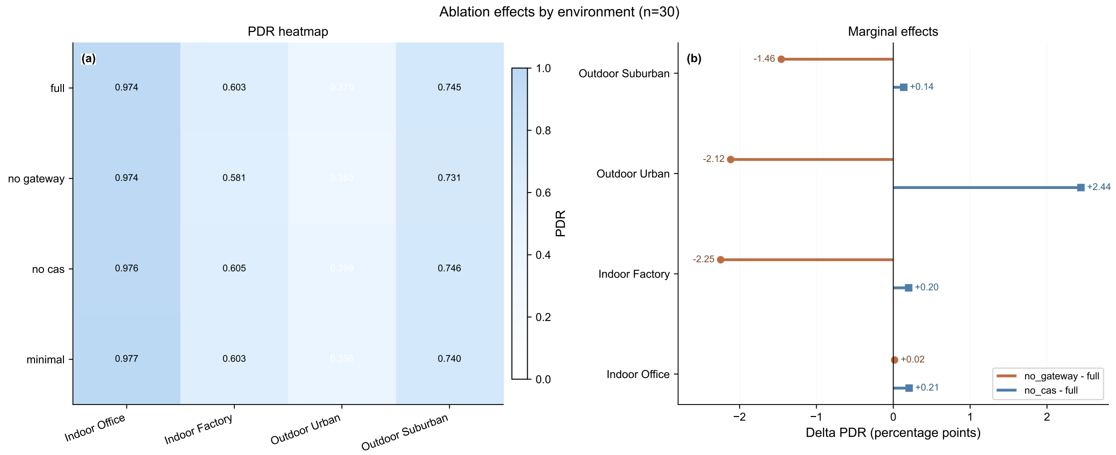

Figure 2. Ablation panel: heatmap and marginal effects (n=30).

5.3. Scalability (100–1000 Nodes; Balanced n=1000 per Cell)

Table 4. PDR ranking at 1000 nodes from the balanced S8 frozen matrix (n=1000 per cell).

Environment	AERIS	LEACH	PEGASIS	HEED	TEEN	AERIS rank
indoor_office	0.9899	0.6282	0.8942	0.8915	0.8032	1st
indoor_factory	0.9726	0.1884	0.1672	0.1634	0.2199	1st
outdoor_urban	0.8846	0.0622	0.0497	0.0341	0.0662	1st
outdoor_suburban	0.9896	0.3048	0.2875	0.3232	0.3669	1st

AERIS is first in all four environments in the current balanced S8 matrix under the reported simulator assumptions. In the same S8 matrix, AERIS also shows a non-physical PDR-scale increase in three environments (indoor_factory, outdoor_urban, outdoor_suburban). A plausible cause is that the original S8 simulator path omits explicit MAC-layer contention penalties, so increasing node density can artificially increase effective forwarding opportunities instead of inducing collision loss. We therefore treat S8 as a frozen baseline regime under the original simulator assumptions and use S9/S10/S11 as calibration evidence for physically stricter interpretation. Interpretation priority in this manuscript is therefore: (i) 100-node reliability ranking, (ii) S8 as baseline scalability snapshot, and (iii) S9/S10/NS-3 as calibration and boundary evidence.

5.4. Statistical Robustness Snapshot (AERIS vs LEACH)

Table 5. Holm-corrected significance snapshot at 1000 nodes (AERIS vs strongest baseline in each environment).

Environment	Nodes	ΔPDR	Hedges' g	Holm p	Significant
indoor_office	1000	+0.095705 (vs PEGASIS)	+4.1500	$< 10^{-300}$	yes
indoor_factory	1000	+0.752703 (vs TEEN)	+138.5282	$< 10^{-300}$	yes
outdoor_urban	1000	+0.818370 (vs TEEN)	+180.0377	$< 10^{-300}$	yes
outdoor_suburban	1000	+0.622649 (vs TEEN)	+98.9577	$< 10^{-300}$	yes

This snapshot highlights strong gains in harsh environments and a moderate but significant gain in indoor_office under the current balanced matrix. For very large sample sizes, practical interpretation in this study still follows both effect size and absolute PDR gap rather than significance labels alone. The very large absolute values of Hedges' g in harsh environments reflect both large mean gaps and low within-group variance under large sample sizes; they should be interpreted as magnitude indicators within this experiment design, not as cross-study constants.[1]

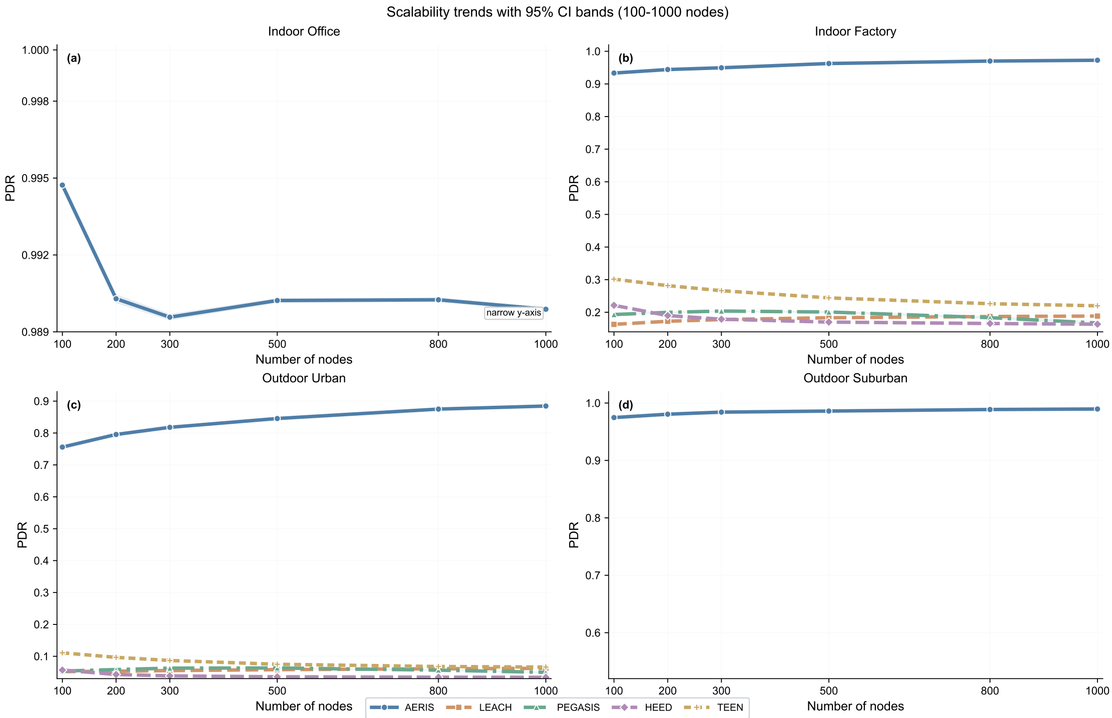

Figure 3. Scalability trends over node counts (balanced n=1000 per environment-node-protocol cell) with 95% CI bands. The indoor_office panel uses a narrowed y-axis window to expose non-zero ordering gaps near PDR=1.0. Under the original simulator settings (without MAC-collision modeling), AERIS exhibits an increasing trend in three environments; this is treated as a known S8 baseline limitation and is cross-checked with the stricter S9/S10 calibration blocks.

5.5. Rigor-Patch Pilot Sanity Check (n=60; not used for core claims)

To test simulator-physics consistency, we ran a publication-tier local pilot with collision and baseline multi-hop relay enabled (`-mac-collision -multihop-relay`), using the same environment taxonomy and nodes {100, 500, 1000} with n=60 per cell. This pilot is a sanity check and does not replace the frozen S8 matrix used for core claims.

Table 6. AERIS PDR under the rigor-patch pilot (n=60 per environment-node cell).

Environment	100 nodes	500 nodes	1000 nodes
indoor_office	0.9706	0.8676	0.6746
indoor_factory	0.9283	0.9083	0.7355
outdoor_urban	0.7487	0.7362	0.1492
outdoor_suburban	0.9563	0.9193	0.7307

The key pilot observation is monotonicity recovery: AERIS PDR decreases with scale in all four environments. In indoor_office, PEGASIS is higher than AERIS under this stricter setting, while AERIS

[1] For large-n comparisons with very small within-group variance, standardized effect sizes can become extremely large. In this manuscript, Hedges' g is therefore interpreted jointly with absolute ΔPDR and confidence intervals, not as a stand-alone cross-study scale.

remains clearly higher in the other three environments. Therefore, this pilot is treated as a gate result: it validates the direction of a simulator-rigor fix, but final paper-level scalability claims remain tied to the frozen matrix until a full rerun is completed under the upgraded simulator settings.

5.6. Matched S9 Patch-vs-Control Stress Test (Four Environments)

To reduce seed and sampling confounders, we ran a matched patch-versus-control stress test in all four environments (indoor_office, indoor_factory, outdoor_urban, and outdoor_suburban). The patch mode enables collision and baseline relay mechanisms; the control mode uses the original simulator path. Patch cells use n=1000 and control cells use n=600.

Table 7. AERIS PDR in matched S9 stress test (patch vs control, four environments).

Environment	Nodes	Patch	Control
indoor_office	100	0.9714	0.9948
indoor_office	500	0.8679	0.9902
indoor_office	1000	0.6779	0.9899
indoor_factory	100	0.9281	0.9329
indoor_factory	500	0.9060	0.9627
indoor_factory	1000	0.7291	0.9726
outdoor_urban	100	0.7482	0.7559
outdoor_urban	500	0.7299	0.8460
outdoor_urban	1000	0.1372	0.8845
outdoor_suburban	100	0.9556	0.9744
outdoor_suburban	500	0.9198	0.9859
outdoor_suburban	1000	0.7280	0.9897

Three observations are robust in the completed S9 evidence: (i) AERIS remains above LEACH/HEED/TEEN under patch mode in all four tested environments at 1000 nodes; (ii) PEGASIS is minimally affected by the patch, with near-zero deltas and non-significant tests in all 24 PEGASIS cells; (iii) AERIS patch-control deltas are significant in all 24 AERIS cells after Holm correction, with the largest absolute drop in outdoor_urban at high scale. In S11, PEGASIS shows exact zero deltas in indoor_factory (all six node counts), which is treated as an implementation-coupling anomaly pending further code-path audit rather than as physical invariance evidence. At the same time, the patch can be overly aggressive for some baseline paths at high scale, so this block is interpreted as a stress-test regime rather than a finalized replacement matrix. To remove the S9 sample-size asymmetry directly, we then run S11 as a matched confirmation block (patch $n = 1000$, control $n = 1000$ per cell) and report that matrix separately below.

5.7. S10 TX-Power Sensitivity (Four Environments)

We additionally evaluate tx-power sensitivity under the upgraded simulation path by comparing 5 dBm versus 15 dBm across four environments, three node counts (100, 500, 1000), and five protocols (n=600 per cell). The full matrix contains 60 matched comparisons (4 environments $\times$ 3 node counts $\times$ 5 protocols), of which 59/60 are significant after Holm correction.

Table 8. AERIS power-sensitivity snapshot at 1000 nodes (S10, metric=`pdr_expected`).

Environment	tx=5 dBm	tx=15 dBm	Delta (5–15)
indoor_office	0.8177	0.5367	+0.2810
indoor_factory	0.6623	0.5051	+0.1571
outdoor_urban	0.0587	0.1606	-0.1019
outdoor_suburban	0.7679	0.5440	+0.2239

The S10 block confirms that power response is environment-dependent and non-monotonic under the current routing and channel settings. In particular, raising tx-power does not guarantee higher PDR in every environment-node cell. The single non-significant comparison in the 60-cell matrix is LEACH at indoor_office, 1000 nodes.

5.8. S11 Matched Patch-vs-Control Matrix (Four Environments, n=1000 vs 1000)

To remove remaining sample-size asymmetry from S9, we completed a fully matched S11 matrix in which both patch and control arms use n=1000 per environment-node-protocol cell (4 environments $\times$ 6 node counts $\times$ 5 protocols). This block directly tests whether stricter physics assumptions improve or degrade reliability under matched sampling.

Table 9. AERIS patch-control deltas in the completed S11 matched matrix (delta = patch - control).

Environment	Delta at 100 nodes	Delta at 1000 nodes
indoor_office	-0.0234	-0.3120
indoor_factory	-0.0052	-0.2435
outdoor_urban	-0.0077	-0.7474
outdoor_suburban	-0.0191	-0.2616

In the S11 matched matrix, AERIS patch-control deltas are negative and significant in all 24 environment-scale cells after Holm correction (24/24 significant; Hedges' g range: -0.47 to -15.36). Therefore, under the current patch configuration (`mac-collision + multihop-relay`), stricter physics assumptions reduce reliability rather than increase it. This means the patch block is evidence for realism stress, not yet evidence for superior calibrated performance.

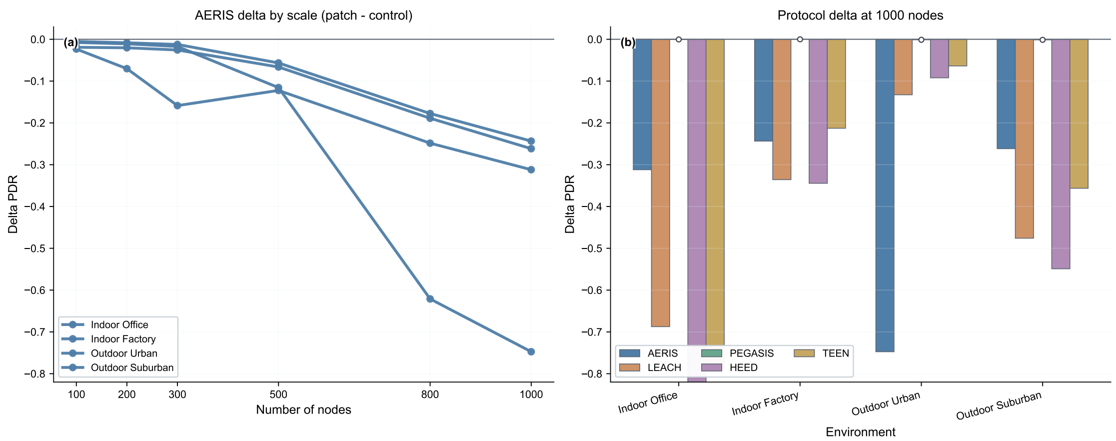

Figure 4. S11 matched patch-control evidence. (a) AERIS delta trajectories over node counts (patch - control). (b) Protocol deltas at 1000 nodes across environments. Negative deltas indicate lower PDR under patch mode.

5.9. Hop-Based Latency and Trade-offs

At 100 nodes (n=30), AERIS stays near two hops (1.97–1.99), while PEGASIS remains above 31 hops due to chain traversal. LEACH and TEEN show lower average hops because of protocol paths that allow direct-to-BS transmissions.

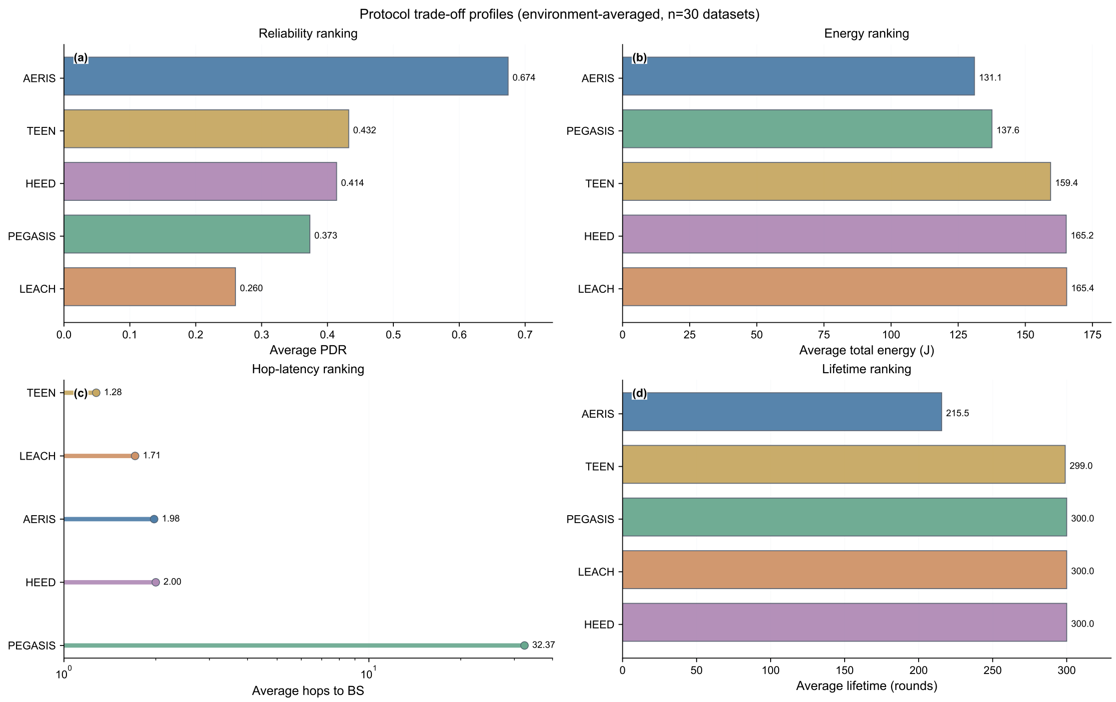

Figure 5. Trade-off panel from publication-tier data: reliability, energy, hop-based latency, and lifetime. The hop panel uses a logarithmic x-axis to keep low-hop and high-hop protocols visible in the same frame without overlap.

5.10. NS-3 Trend Validation (AERIS vs LEACH, n=30)

NS-3 validation is used as a cross-platform trend check rather than a numerical-equivalence proof. This block is restricted to AERIS-versus-LEACH and is not used for five-protocol cross-platform ranking. Under aligned multi-environment settings, AERIS is directionally not lower than LEACH in 27 of 28 tested environment-scale cells; the single exception is indoor_office at 200 nodes (AERIS–LEACH diff $= -0.0004$, not significant). At 100 nodes, significance is confirmed in 3/4 environments (indoor_factory, outdoor_urban, outdoor_suburban), while indoor_office remains non-significant. In the 300/500-node extension, all eight environment-scale comparisons are significant after Holm correction; with the 800/1000 extension included, non-significance remains confined to indoor_office at 1000 nodes.

Table 10. NS-3 trend-level validation at 100 nodes (AERIS vs LEACH, n=30; metric=`pdr_expected`).

Environment	AERIS	LEACH	Delta	Holm-corrected p	Significance
indoor_office	0.9202	0.9185	+0.0017	1.00	no
indoor_factory	0.6025	0.5336	+0.0690	4.23×10^{-25}	yes
outdoor_urban	0.2064	0.1899	+0.0165	3.41×10^{-3}	yes
outdoor_suburban	0.7771	0.6921	+0.0850	1.94×10^{-30}	yes

Scope note: this NS-3 block supports trend consistency only. The manuscript does not claim cross-platform numerical equivalence. Across 7 node counts and 4 environments, 25/28 AERIS-versus-LEACH comparisons are significant; the non-significant cells are indoor_office at node counts 100, 200, and 1000.

6. Discussion

The evidence supports environment-scoped positioning rather than universal superiority. In the reported matrices, AERIS is strongest on reliability in harsh channels.

The manuscript separates three evidence blocks:

- 100-node multi-environment comparison,
- large-scale regime (100–1000 nodes) with balanced n=1000 per cell,
- NS-3 trend validation for AERIS versus LEACH only.

Cross-block numerical pooling is avoided because these blocks answer different questions. Under stricter S9/S11 collision-relay settings, high-scale PDR decreases for most protocols and PEGASIS remains comparatively stable. Under S10, tx-power effects are strong but not globally monotonic. Practical interpretation therefore remains environment-scoped.

Practical interpretation also requires separating statistical and engineering significance. At large sample sizes, very small absolute differences can be significant after correction. For this reason, we interpret each result jointly by corrected p value, effect size, and absolute PDR delta. Energy interpretation is similarly conditioned: lower total energy can co-occur with shorter lifetime when earlier network termination occurs.

Practical Deployment Guidance

Based on the completed matrices, the following deployment guidance is supported by the reported evidence:

- In harsh environments (indoor_factory, outdoor_urban, outdoor_suburban), enabling Gateway support is recommended because no-gateway ablations reduce reliability.
- In benign indoor_office conditions, gains are smaller and model overhead should be justified against application reliability targets.
- For high-scale deployments, treat S8 as baseline evidence and use S9/S10/S11 calibration before final engineering parameter selection.
- Tx-power settings should be environment-tuned; monotonic power assumptions are not supported by S10.

Table 11. Environment-scoped engineering interpretation from the reported evidence.

Environment type	Recommended interpretation	Main caveat
indoor_office	Reliability differences are small at scale; optimize for deployment cost first	Several cells are non-significant in NS-3 trend checks
indoor_factory	AERIS/Gateway gains are most stable and practically relevant	Validate tx-power and collision settings before deployment
outdoor_urban	AERIS remains preferable under harsh links	High-scale behavior is sensitive to physics assumptions
outdoor_suburban	AERIS trend remains favorable in both simulator and NS-3 trend blocks	Avoid monotonic tx-power assumptions

Reviewer-Critical Validity Notes

Three validity boundaries are explicit in this draft. First, the reported comparisons are simulator-based under the tested channel taxonomy. Second, NS-3 evidence is trend-level only. Third, module contributions are context dependent: Gateway and CAS effects vary by environment and should not be interpreted as globally monotonic benefits.

7. Limitations and Future Work

- Current evidence is simulator-based.
- NS-3 trend-level publication gate is completed; numerical equivalence is outside the current validation scope.
- NS-3 validation in this manuscript focuses on AERIS versus LEACH; full five-protocol cross-platform ranking remains future work.

- The large-scale S8 matrix is balanced at n=1000 per cell and should be interpreted under the original simulator assumptions.
- A matched S9 patch-versus-control stress test has now been completed in all four environments (patch n=1000, control n=600 per environment-node-protocol cell); unequal sample sizes come from phased execution and are handled with Welch's test.
- A fully matched S11 patch-versus-control matrix has also been completed (patch n=1000, control n=1000 per environment-node-protocol cell); AERIS deltas are negative and significant in all 24 environment-scale cells under the current patch configuration.
- A matched S10 tx-power sensitivity matrix has now been completed in all four environments (tx=5 dBm vs 15 dBm, n=600 per environment-node-protocol cell); 59/60 cells are significant, but response direction is environment-dependent.
- Under patch mode, high-scale baseline PDR can become very low in benign channels, so collision/relay parameter calibration is still required before replacing the frozen S8 matrix in the final camera-ready version.
- Reported gains are limited to the tested baseline set and channel taxonomy.
- Hop count is reported as a route-length proxy, not as a wall-clock latency benchmark under full MAC scheduling.
- Additional topology families (corridor, hotspot) and hardware-in-the-loop validation remain pending.

8. Conclusion

AERIS provides a reliability-oriented routing option for heterogeneous WSN channels. In the publication-tier 100-node matrix, it achieves the highest mean PDR within the tested baseline set across all four environments. In the balanced S8 large-scale matrix (100–1000 nodes, n=1000 per cell), AERIS is the highest-mean protocol in all tested environments; however, S8 is treated as baseline evidence because it excludes MAC-collision modeling and shows non-physical upward trends in three environments. NS-3 adds trend-level support for AERIS versus LEACH. The completed S9 stress block, S10 power-sensitivity matrix, and S11 fully matched patch-control matrix show that stricter physics and power changes can materially alter high-scale behavior; in S11, AERIS patch-control deltas are negative and significant in all 24 environment-scale cells under the current patch setup. Collision/relay parameter calibration is therefore required before replacing S8 with a finalized camera-ready scalability matrix.

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Conflicts of Interest: The authors declare no conflict of interest.

Data Availability Statement

The datasets, scripts, manuscript sources, figure-generation code, and provenance sidecar records (including commit identifiers and script hashes) supporting this study are available from the corresponding author and will be released in the public project repository upon final publication. The released package includes five evidence groups: 100-node multi-environment comparison, frozen S8 scalability summary and significance tables, S9 and S11 patch-control calibration tables, and the NS-3 trend-level gate summary.

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